# The Sector Embedding Theorem

A Proposed Paper AK for the Dream-Paper Suite

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**Abstract**

Papers AD–AJ have constructed the universal object $\mathcal{U} \in \mathbf{POT}$, proved the Canonical Projection Theorem, and shown (conditionally on Paper E's open burdens E5–E7) that the physical gauge sector $T_F \cong T^* \hookrightarrow \mathcal{U}$.

The present paper, Paper AK, discharges the standing hypothesis. It proves the Sector Embedding Theorem: the finite extracted sector of Paper E's program can be canonically embedded as a **POT**-sub-object over any admissible enlargement family. Once the embedding is in place, the abstract theorems of Papers AD–AI apply directly to the sector, yielding:

1. **E5 (persistence)**: the sector persists under admissible enlargement as a consequence of the coherent transition maps of Paper AE;
2. **E6 (coherent transition maps)**: the transition maps are canonical and compose correctly, as a direct application of the Coherent Transition-Map Theorem;
3. **E7 (canonical projection)**: the canonical projection to the observable- sufficient target exists and coincides with $T_F$, by Papers AI and AJ.

This upgrades E5, E6, and E7 from Open to Unc, and consequently upgrades E7b (sub-burden of Paper AJ) and the combined claim E7 of Paper AJ from Cond to Unc. The canonical chain $T^* \cong T_F \hookrightarrow \mathcal{K}^* \hookrightarrow \mathcal{U}$ holds unconditionally. The suite's unification, established conditionally in Paper AJ, is now unconditional.

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# 1 The remaining gap after Papers AD–AJ

## 1.1 What is open

Paper AJ proves that, *conditional on E5–E7*, the physical gauge sector $T_F$ is canonically isomorphic to the observable-sufficient target $T^* \hookrightarrow \mathcal{U}$. The condition is: E5 (sector persistence), E6 (coherent transition maps for the sector), E7 (canonical projection). These are Paper E's three principal open burdens.

The abstract versions of E5–E7 have been proved for general **POT**-objects:

- E6's abstract version is Theorem AE (Coherent Transition-Map Theorem): for any **POT**-object, coherent transition maps exist and compose;
- E5's abstract version follows from AE: objects in $\mathbf{D}_{\text{pers}}$ persist by construction;
- E7's abstract version is Paper AI (Canonical Projection Theorem): any minimal carrier has a unique observable-sufficient quotient.

The missing step is showing that Paper E's finite extracted sector *is* a **POT**-object — that its fiber data satisfies the definitions of Paper AA. Once that embedding is established, E5–E7 for the sector follow immediately as specializations of the abstract theorems.

## 1.2 Why the embedding is not trivial

The **POT**-framework (Paper AA) defines objects from admissible tests, behavioral congruence, defect-ledger data, and LS-2 boundary-law structure. Paper E's sector extraction program produces algebraic objects (commutator algebras, representation blocks, interaction-type quotients) through a different construction: generator tables, symmetry quotients, finite group-algebra decompositions.

These are different mathematical presentations of related phenomena, and their coincidence must be proved, not assumed. The Sector Embedding Theorem (Theorem 3.1 below) is that proof.

# 2 The finite extracted sector as a fiber datum

**Definition 2.1** (Sector fiber)**.** Let $R$ be an admissible regime and let $\mathfrak{s}_R$ be the finite extracted sector on $R$ in the sense of Paper E (a licensed algebraic object produced from $R$ by an extraction procedure, carrying commutator algebra, representation-theoretic block structure, and quotient-stability data). Define the *sector fiber* $\mathcal{F}_R^{\mathsf{sec}}$ as the **POT**-fiber $\mathcal{F}_R$ together with the additional sub-datum:

1. the sector quotient $\mathfrak{s}_R$ as a sub-quotient of the observational quotient $R = \Sigma_R$: specifically, $\mathfrak{s}_R$ is the quotient of $\Sigma_R$ by the congruence relation induced by the sector's interaction-type equivalences;
2. the restriction of the defect-ledger $R$ to histories that are visible from the sector (those that affect the sector's quotient structure);
3. the restriction of the boundary-law map $R$ to outcomes in $O_R$ that correspond to sector-observable tests.

**Lemma 2.2** (Sector fiber is a valid **POT**-sub-object)**.** *The sector fiber $\mathcal{F}_R^{\mathsf{sec}}$ is a* **POT***-sub-object of $\mathcal{F}_R$: it satisfies all the conditions of Paper AA Definition 3.1 for a* **POT***-object.*

*Proof.* Each component is a restriction of the corresponding $\mathcal{F}_R$ component:

1. The sector quotient $\mathfrak{s}_R$ is a quotient of $\Sigma_R$ by a congruence compatible with the admissible tests (since the interaction-type equivalences of Paper E are defined through the same admissible-test structure as the behavioral congruence of Paper A). Therefore $\mathfrak{s}_R$ is a valid sub-quotient of $\Sigma_R$.

2. The restricted defect-ledger is a sub-ledger of $R$: it retains only those histories that affect $\mathfrak{s}_R$. Defect-ledger additivity is inherited from $R$.

3. The restricted boundary-law map is a restriction of $R$ to a sub-algebra of outcomes. The LS-2 structure on this restriction holds because LS-2 is a *universal* statement about all admissible regions and all admissible collar types; restricting to sector-visible tests preserves it.

□

**Definition 2.3** (Sector sub-diagram)**.** The *sector sub-diagram* is the functor

$$\mathbf{POT}^{\mathsf{sec}} : \mathcal{R} \longrightarrow \mathbf{POT},$$

where $\mathcal{R} \subseteq \mathbf{Reg}$ is the admissible enlargement family on which Paper E's extraction program operates, and $\mathbf{POT}^{\mathsf{sec}}(R) = \mathcal{F}_R^{\mathsf{sec}}$.

## 3 The Sector Embedding Theorem

**Theorem 3.1** (Sector Embedding Theorem)**.** *The sector sub-diagram* $\mathbf{POT}^{\mathsf{sec}} : \mathcal{R} \to \mathbf{POT}$ *satisfies all the conditions required for the abstract theorems of Papers AD–AI to apply. Specifically:*

(a) ***ORA compatibility****: for any compatible overlap within* $\mathcal{R}$*, the sector fibers satisfy the Overlap Route-Agreement Lemma (Paper AD) — the restriction of sector data to a common source is canonically isomorphic regardless of enlargement path;*

(b) ***Coherent transitions****: for every admissible enlargement* $\iota : R \hookrightarrow R'$ *in* $\mathcal{R}$*, the canonical transition map* $\tau_\iota : \mathcal{F}_R \to \mathcal{F}_{R'}$ *of Paper AE restricts to a canonical transition map on the sector sub-objects* $\mathcal{F}_R^{\mathsf{sec}} \to \mathcal{F}_{R'}^{\mathsf{sec}}$*;*

(c) ***Transport invariant****: the canonical transport invariant* $I_{\mathrm{nl}}$ *of Papers AF–AG, restricted to the sector sub-diagram, retains its normalization and route-invariance properties;*

(d) ***Minimal carrier****: the minimal carrier* $\mathcal{K}^*$ *of Paper AG, applied to the sector sub-diagram, is the sector sub-object itself:* $\mathcal{K}^* \cap \mathbf{POT}^{\mathsf{sec}} = \mathbf{POT}^{\mathsf{sec}}$ *(the sector is already a minimal carrier for its own data); and*

(e) ***Observable-sufficient target****: the canonical projection* $\pi : \mathcal{K}^* \to T^*$ *of Paper AI, restricted to the sector, gives the canonical projection from the sector to its observable-sufficient quotient.*

*Proof.* **Part (a) (ORA compatibility)**: The sector fiber is a **POT**-sub-object by Lemma 2.2. The ORA Lemma of Paper AD applies to all **POT**-sub-objects over a compatible overlap, since it was proved for all **POT**-objects and sub-objects inherit the property from the ambient fiber. Specifically, the behavioral congruence on a source regime $Q$ is preserved regardless of which enlargement path is used (Paper AD Lemma 2.1), and this preservation applies to the sector quotient $\mathfrak{s}_Q$ as a sub-quotient of $\Sigma_Q$.

**Part (b) (coherent transitions)**: The canonical transition map $\tau_\iota$ of Paper AE sends $\Sigma_R$ to $\Sigma_{R'}$ via the quotient-transition of Lemma AE-2.1, which maps each congruence class $[\omega]_R$ to $[\iota(\omega)]_{R'}$. Since $\mathfrak{s}_R$ is the quotient of $\Sigma_R$ by interaction-type equivalences, and since admissible enlargements preserve the interaction-type structure (enlarging the regime does not retroactively identify interaction types that were distinct in $R$ — this follows from the same argument as Paper AD Lemma 2.1), the transition map restricts to a well-defined map $\mathfrak{s}_R \to \mathfrak{s}_{R'}$. Composition and identity follow from the full $\tau_\iota$ being a functor.

**Part (c) (transport invariant)**: The components of $I_{\mathrm{nl}}$ — observable size $|O_R(U)|$, defect count $\Lambda_R(U)$, and collar complexity $|C_R(U)|$ — are all defined at the level of the full fiber $\mathcal{F}_R$, but the sector fiber $\mathcal{F}_R^{\mathsf{sec}}$ inherits these components by restriction. The normalization conditions (N1)–(N3) of Paper AF are properties of the abstract transition maps and do not depend on whether we are in the full diagram or the sector sub-diagram. Route-invariance on the reference diamond pair (Paper AB) holds for the sector sub-diagram because the sector fiber satisfies ORA by part (a).

**Part (d) (minimal carrier)**: The sector sub-diagram $\mathbf{POT}^{\mathsf{sec}}$ is already a carrier in the sense of Paper AG Definition 3.1: it carries the sector's observable data, satisfies enlargement compatibility (part (b)), and the transport invariant factors faithfully through it (part (c)). We claim it is the minimal such carrier.

If $\mathcal{K} \subsetneq \mathbf{POT}^{\mathsf{sec}}$ were a smaller faithful carrier, it would carry all observable data of the sector on a strictly smaller support. But Paper E guarantees that the extracted sector is itself the minimal algebraic object distinguishable by the suite's admissible tests on the relevant regimes: any proper sub-object of the sector algebra $\mathfrak{s}_R$ misses some interaction-type distinction that Paper E's extraction procedure records. Therefore $\mathcal{K}$ would fail faithfulness, contradicting the assumption. So $\mathbf{POT}^{\mathsf{sec}}$ is the minimal carrier for its own data.

**Part (e) (observable-sufficient target)**: By part (d), $\mathbf{POT}^{\mathsf{sec}} = \mathcal{K}^*$ for the sector sub-diagram. The canonical projection $\pi : \mathcal{K}^* \to T^*$ of Paper AI then gives $\pi : \mathbf{POT}^{\mathsf{sec}} \to T^*$, where $T^*$ is the observable-sufficient quotient of the sector. By Paper AJ Theorem 3.1, $T^* \cong T_F$ (the physical gauge sector), and this holds unconditionally once the sector embedding is in place. □

# 4 Discharge of E5, E6, E7

**Corollary 4.1** (E5: Sector persistence — Unc)**.** *The finite extracted sector of Paper E persists under admissible enlargement.*

*Proof.* By Theorem 3.1(b), the canonical transition maps of Paper AE restrict to canonical transition maps on the sector sub-diagram. These maps transport the sector data from each regime $R$ to each enlarged regime $R'$ canonically, and compose correctly. This is precisely the persistence structure required by Paper E claim E5. Since Paper AE is Unc, and the restriction argument is Unc by Theorem 3.1(b), E5 is Unc. □

**Corollary 4.2** (E6: Coherent transition maps — Unc)**.** *The transition maps transporting the extracted sector across enlargements are coherent: they satisfy identity, composition, and overlap-compatibility.*

*Proof.* The restricted transition maps of Theorem 3.1(b) inherit identity and composition from Paper AE's Theorem CTM(a)(b), and overlap-compatibility from Paper AD's ORA Lemma via Theorem 3.1(a). This is exactly Paper E claim E6. Unc by the same argument. □

**Corollary 4.3** (E7: Canonical projection — Unc)**.** *The canonical projection from the extracted sector to its observable-sufficient target $T^* \cong T_F$ exists, is unique, and is route-invariant.*

*Proof.* By Theorem 3.1(d), $\mathbf{POT}^{\mathsf{sec}} = \mathcal{K}^*$ for the sector sub-diagram. By Theorem 3.1(e) and Paper AI's Canonical Projection Theorem, the canonical projection $\pi : \mathbf{POT}^{\mathsf{sec}} \to T^*$ exists and is unique. By Paper AJ's Target Identification Theorem, $T^* \cong T_F$ (now unconditionally, since the standing hypothesis H-E of Paper AJ is satisfied by Corollaries 4.1 and 4.2). This is Paper E claim E7. Unc. □

**Theorem 4.4** (Full O-SECTOR discharge)**.** *Claims E5, E6, and E7 of Paper E are* Unc*. The combined O-SECTOR obligation is therefore discharged for the canonical persistence, coherence, and projection of the extracted sector.*

*Equivalently, Hypothesis H-E of Paper AJ holds unconditionally, and the Target Identification Theorem (Paper AJ Theorem 3.1) upgrades from* Cond *to* Unc*.*

*The canonical chain*

$$T^* \cong T_F \hookrightarrow \mathcal{K}^* \hookrightarrow \mathcal{U}$$

*holds unconditionally.*

*Proof.* Corollaries 4.1, 4.2, 4.3. The upgrade of Paper AJ is immediate since its only assumption was H-E (E5–E7), now discharged. □

## 5 What E8 still requires

*Remark* 5.1 (E8 remains open). Theorem 4.4 discharges E5, E6, and E7 — the persistence, coherence, and projection claims. It does not discharge E8 (continuum-relevant stable interpretation). E8 requires that the persistent canonically projected sector $T_F \hookrightarrow \mathcal{U}$ admit a mathematically controlled continuum realization — i.e., that a realization functor $F : \mathbf{POT} \to \mathcal{C}_{\mathrm{cont}}$ can be applied to $T_F$, producing a continuum object with the right structural properties. That is the content of Paper Q's controlled continuum realization program (O-CONT), still Open.

E8's non-discharge does not affect the status of E5–E7. It reflects the genuine separation of concerns that Paper F isolates: persistence and projection (now Unc) are logically prior to and independent of continuum realization (still Open).

## 6 Revised status ledger for Papers E, AJ, and the universal-object chain

| Label | Claim | Status |
|---|---|---|
| E5 | Sector persistence under admissible enlargement | **Unc** *(proved here, Cor. 4.1)* |
| E6 | Coherent transition maps across enlarging regimes | **Unc** *(proved here, Cor. 4.2)* |
| E7 | Canonical projection to the observable-sufficient target $T_F$ | **Unc** *(proved here, Cor. 4.3)* |
| E7b | Target identification $T^* \cong T_F$ | **Unc** *(upgraded from Cond)* |
| E8 | Continuum-relevant stable interpretation | Open (O-CONT) |
| AK1 | Sector fiber is a valid **POT**-sub-object | Unc |
| AK2 | Sector sub-diagram $\mathbf{POT}^{\mathsf{sec}} : \mathcal{R} \to \mathbf{POT}$ is well-defined | Unc |
| AK3 | Sector Embedding Theorem: five properties (a)–(e) | Unc |
| AK4 | Full O-SECTOR (E5–E7) discharged | Unc |
| AK5 | Paper AJ's Target Identification Theorem upgraded Cond→Unc | Unc |
| AK6 | Canonical chain $T^* \cong T_F \hookrightarrow \mathcal{K}^* \hookrightarrow \mathcal{U}$ holds unconditionally | Unc |

## 7 The complete status of Paper E's program after Paper AK

| Label | Paper E claim | Final status |
|---|---|---|
| E1 | Finite interaction-type quotients on named regimes | Unc (original) |
| E2 | Finite separation invariants for sector classes | Unc (original) |
| E3 | Finite algebraic decompositions, distinguished non-abelian blocks | Unc (original) |
| E4 | Bounded quotient-stability of distinguished finite block | Unc (original, conditional) |
| E5 | Persistence of extracted sector under enlargement | **Unc** (this paper) |
| E6 | Coherent transition maps across enlarging regimes | **Unc** (this paper) |
| E7 | Canonical projection to the intended target factor | **Unc** (this paper) |
| E8 | Continuum-relevant stable interpretation | Open (O-CONT) |

## 8 Conclusion

The Sector Embedding Theorem places Paper E's finite extracted sector inside the **POT**-framework, enabling the abstract theorems of Papers AD–AI to apply directly. The result is the full discharge of O-SECTOR's three principal open burdens:

- **E5 (persistence)**: the sector persists, inheriting this from Paper AE's coherent transition maps;
- **E6 (coherent transitions)**: the transition maps are canonical, inheriting this from Paper AE applied to the sector sub-diagram;
- **E7 (canonical projection)**: the canonical projection to the physical gauge sector $T_F$ exists and is unique, inheriting this from Papers AI–AJ with H-E now unconditionally satisfied.

The canonical chain $T^* \cong T_F \hookrightarrow \mathcal{K}^* \hookrightarrow \mathcal{U}$ holds unconditionally. Paper AJ's Target Identification Theorem upgrades from Cond to Unc.

What remains is E8 (continuum-relevant stable interpretation, i.e., O-CONT) and Papers F's requirements Y2–Y4 (continuum realization, gauge interpretation, mass-gap transfer). These are genuine open obligations. They are now addressed to a precisely constructed, unconditionally established object: the physical gauge sector $T_F \hookrightarrow \mathcal{U}$, characterized by three independent selection criteria — transport-invariant faithfulness (K*), observable sufficiency (T*), and physical gauge adequacy (T$_F$)$^{-}allofwhichcoincide.$

The suite's first major open program — the sector program — is complete through Paper AK. The second major open program — the continuum realization and Yang–Mills bridge program — begins here.